

Storage Classes in C

How variables are stored, used, and shared

What is a Storage Class?

In C, a **storage class** is a keyword that determines the **scope, lifetime, visibility, and memory location** of a variable or function.

While a data type tells the compiler how much memory to allocate, the storage class tells it **how to manage** that memory and where the variable can be accessed.

What is a Storage Class?

It defines 4 things about a variable:

1. Scope → Where it can be accessed
2. Lifetime → How long it exists in memory
3. Default Value → The value automatically assigned to a variable if it is not explicitly initialized.
4. Storage Location (Memory or CPU register)

Types of Storage Classes in C

1. auto
2. register
3. static
4. extern

auto (Default)

- Created inside function (local variable)
- Scope: Inside function only
- Lifetime: Till function ends
- Default value: Garbage

Example:

```
int main()
{
    auto int x;
    printf("%d", x);
}
```

register

- Stored in CPU register → very fast access
- Used for loop counters
- Registers do not have standard memory addresses, you cannot use the address-of (&) operator on a register variable.

Example:

```
int main()
{
    register int i;
    for(i=1; i<=3; i++)
        printf("%d ", i);
}
```

static

- Keeps its value even after function ends
- **Scope:** Inside function only
- **Lifetime:** Entire program

Example:

```
#include<stdio.h>
void fun();
int main()
{
    fun();
    fun();
    fun();
}
void fun() {
    static int x = 0;
    x++;
    printf("%d ", x);
}
```

extern

- Used to share variable between files
- Defined in one file, used in another

File1:

```
int x = 10;  
void fun();  
int main(){ fun(); }
```

File2:

```
extern int x;  
void fun(){ printf("x = %d", x); }
```


Comparison Table

Comparison of the 4 Storage Classes in C

Storage Class	Keyword	Storage Location	Scope	Lifetime	Default Value
Automatic	auto	Stack (RAM)	Local (Block)	Until block exits	Garbage value
Register	register	CPU Register	Local (Block)	Until block exits	Garbage value
Static	static	Data Segment (RAM)	Local or File	Program end	Zero (0)
External	extern	Data Segment (RAM)	Global (All files)	Program end	Zero (0)

Summary

- auto → Default local variable
- register → Fast storage in CPU
- static → Remembers value across calls
- extern → Shares variable across files